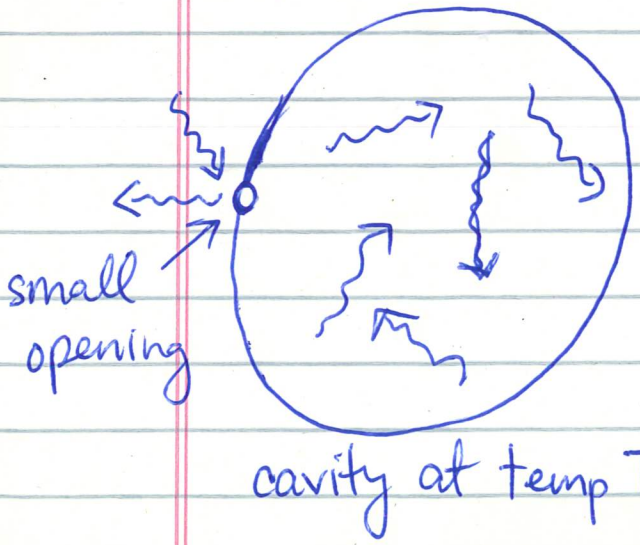
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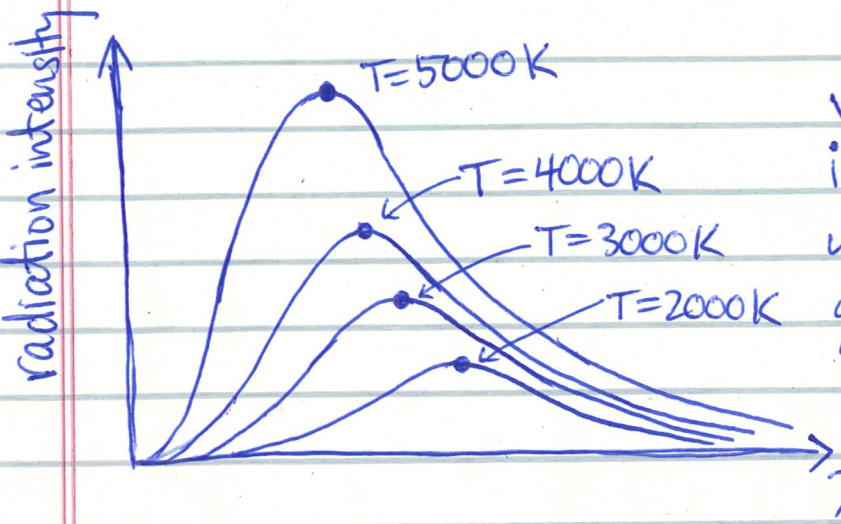
We will now focus for the remainder of the course on key concepts and phenomena from quantum mechanics. For reasons that we will elucidate as we go, this subject requires you to abandon all your intuitions about how reality behaves and build up new intuitions about the quantum world. Before we get there, let's spend today talking about the first pieces of evidence that there was more to physics than the classical description.

① Blackbody radiation

It was well known that an object glows when it is heated (hot stove element, lightbulb filament, etc.) and that the spectrum of light emitted from these objects all follow a similar pattern called a blackbody spectrum. Real objects are not quite a perfect blackbody but they are close

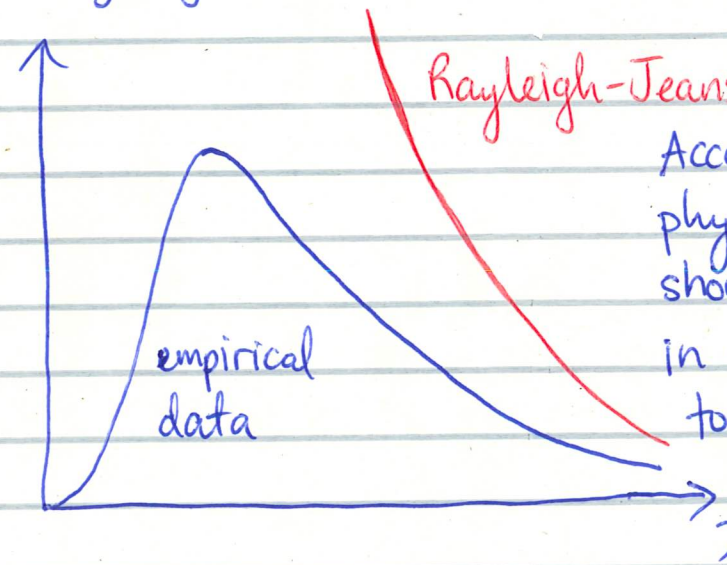


A perfect blackbody is a perfect absorber and a perfect emitter of radiation when in thermal equilibrium. Light enters and leaves through small hole in the cavity wall.



vertical axis is the intensity radiated per unit wavelength. This graph is the empirical data and the peak corresponds to temp. T

The problem: treating the light inside the cavity as a series of resonant modes similar to a guitar string, the classical prediction worked out by Rayleigh and Jeans had some fatal flaws:



According to classical statistical physics, each resonant mode should carry the same energy in thermal equilibrium, leading to a runaway effect at small wavelengths.

The classical calculation predicted nonsense results. Max Planck tried something radically different: he treated the walls of the cavity as if it were a bunch of tiny harmonic oscillators that can only absorb and emit light in packets of energy

$$E_n = nhf \qquad n = 1, 2, 3, \ldots$$

where h is Planck's constant:

$$h = 6.626 \times 10^{-34}\ \text{J}\cdot\text{s} = 4.136 \times 10^{-15}\ \text{eV}\cdot\text{s}$$

The idea is that these atomic oscillators absorb and emit at very specific frequencies instead of continuously. This meant the light itself was <u>quantized</u> into energy packets that Planck called quanta.

The result: $$\boxed{I(\lambda, T) = \frac{2\pi h c^2}{\lambda^5} \frac{1}{e^{hc/\lambda k_B T} - 1}}$$

which matches the blackbody curve exactly and resolves the Rayleigh-Jeans ultraviolet catastrophe.
Here: $c = 3.00 \times 10^8$ m/s is speed of light
$k_B = 1.38 \times 10^{-23}$ J/K is Boltzmann's constant
T is the temperature in Kelvin

From this, two more useful laws can be derived:

<u>Wien's displacement law:</u> $$\boxed{\lambda_{max} T = 2.898 \times 10^{-3}\ \text{m}\cdot\text{K}}$$

relates the peak wavelength of the emission intensity to the temperature of the black body.:

$\lambda_{max} = \frac{\text{constant}}{T}$ $\Rightarrow$ smaller peak wavelength means higher temperature and vice versa

Stefan's Law: $\boxed{P(T) = \sigma A T^4}$ (power radiated)

where $\sigma = 5.670 \times 10^{-8}\ W/(m^2 \cdot K^4)$ is Stefan-Boltzmann const.
A is the surface area of the blackbody
T is the temperature of the blackbody.

Example: The sun's peak wavelength is 500 nm and the Sun has a radius is 6.957×10^5 km. What is the power output of the Sun?

Solution: We have σ and $A = 4\pi r^2$ but we need T in order to calculate the power using Stefan's Law. We also have the peak wavelength, so we can get T using Wien's displacement law:

$$T = \frac{2.898 \times 10^{-3}\ m \cdot K}{500 \times 10^{-9}\ m} = 5800\ K$$

Now we use Stefan's Law:

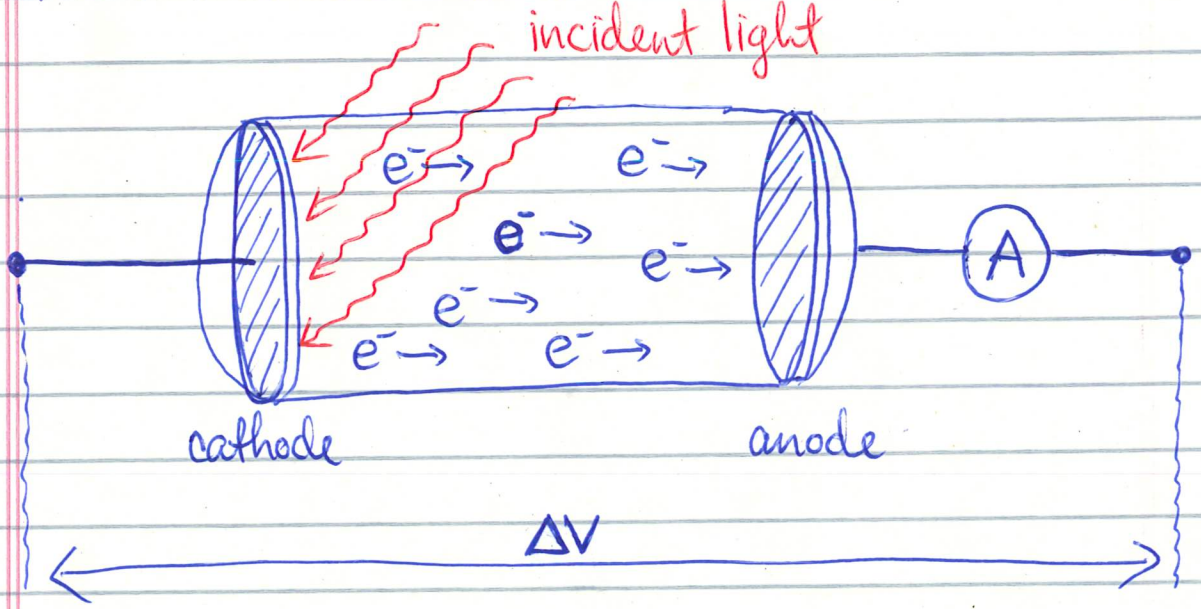
$$P(5800\ K) = (5.67 \times 10^{-8}\ W/(m^2 \cdot K^4))\, 4\pi (6.957 \times 10^8\ m)^2 (5800\ K)^4$$
$$= \boxed{3.90 \times 10^{26}\ W}$$

By comparison, an atomic bomb releases $\sim 2 \times 10^{17}$ J so this is like 4 billion atomic bombs going off every second.

5-minute break

The revolutionary idea that light exists in quantized packets of energy from Planck's resolution of the blackbody problem was the first hint at quantum mechanics. However, Planck himself only thought this was a neat trick to solve the problem and nobody else took the idea seriously except for Einstein. There was another puzzling phenomenon that classical physics could not explain that Einstein was able to resolve using Planck's idea.

② The Photoelectric Effect.



An evacuated tube has a cathode and an anode, light is shone on the cathode and a current is picked up by the anode as electrons are

liberated from the cathode. A potential difference is established externally. Here is what is observed:

- the photocurrent requires frequency of light above a certain threshold $f > f_{crit}$.
- If the anode is at higher potential than cathode, photocurrent is detected $\Rightarrow$ electrons are being liberated from the metal
- If the anode is at a lower potential than cathode, there is a stopping voltage where current no longer flows.
- For low intensity light, there is no lag time b/w light hitting the cathode and observing a photocurrent.

Why can classical physics not explain this?

Clearly electrons are being liberated from the metal, which takes some minimum energy to overcome (binding energy). If light were a classical wave then the energy carried by the wave is related to the square of the amplitude, so the frequency should not matter at all. Classically a photocurrent should flow at any frequency provided the intensity is high enough to allow the electrons to gain sufficient energy to be liberated from the metal. This also implies there would be a lag time as the electrons continuously absorb light energy until

they are freed.

Einstein's picture: light is composed of quanta with energy $E = hf$ and these individual quanta are absorbed by the electrons in the cathode.

Consequences:

① If it takes an energy ϕ (aka the work function) to liberate an electron from the metal, there is a minimum frequency of light that can achieve this:

$$hf_{crit} = \phi \quad \text{(emitted } e^- \text{ is at rest)}$$

② Above this frequency the electron is emitted with some kinetic energy:

$$hf = \phi + K_{max}$$

K_{max} = kinetic energy after leaving metal

$$\Rightarrow \boxed{K_{max} = hf - \phi}$$

also covers point ① above and requires $K_{max} \geq 0$

③ There is no lag time b/w e^- absorbing a photon and being released from the metal.